# Does Twitter feel the war in real time?

VADER sentiment vs. major Russia-Ukraine events at CHTC scale

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STAT 405 / DSCP · April 2026

## Introduction

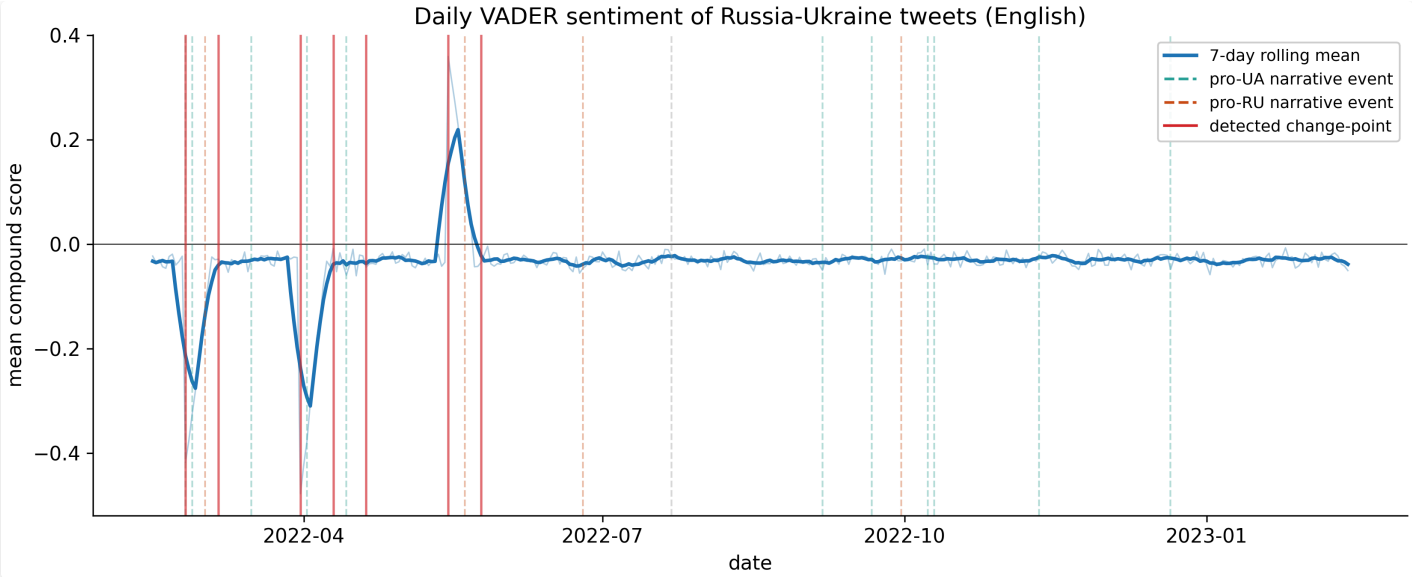
We asked whether the sentiment of public tweets shifts when a major Russia-Ukraine war event happens, how sharply, and how quickly it decays. We scored ≈20 GB of tweets from the Kaggle `bwandowando/ukraine-russian-crisis-twitter-dataset-1-2-m-rows` on CHTC with one HTCondor job per daily CSV (≈200 parallel jobs), then ran three analyses against a curated timeline of 24 events: Welch event-study *t*-tests, PELT change-point detection on the daily mean, and an OLS regression of daily sentiment on event dummies with controls. **Three events show a sentiment shift of ≥0.15 compound units ($|t| > 2.5$)**: the invasion, Azovstal's surrender, and Kherson's liberation; the change-point detector finds the first two without being told.

## Data, computation, and results

**Data.** The dataset comprises ≈200 daily CSV files (50 MB-2 GB each; ≈20 GB total), Feb 2022 - mid-2024. We used five columns: `text`, `tweetcreatedts`, `language`, `retweetcount`, `followers`. Cleaning was minimal: rows with empty text or an unparseable timestamp were dropped (≈1.8% total). We kept every language; primary results pool only `language=="en"` because VADER is English-trained, with a multilingual pool reported as sensitivity.

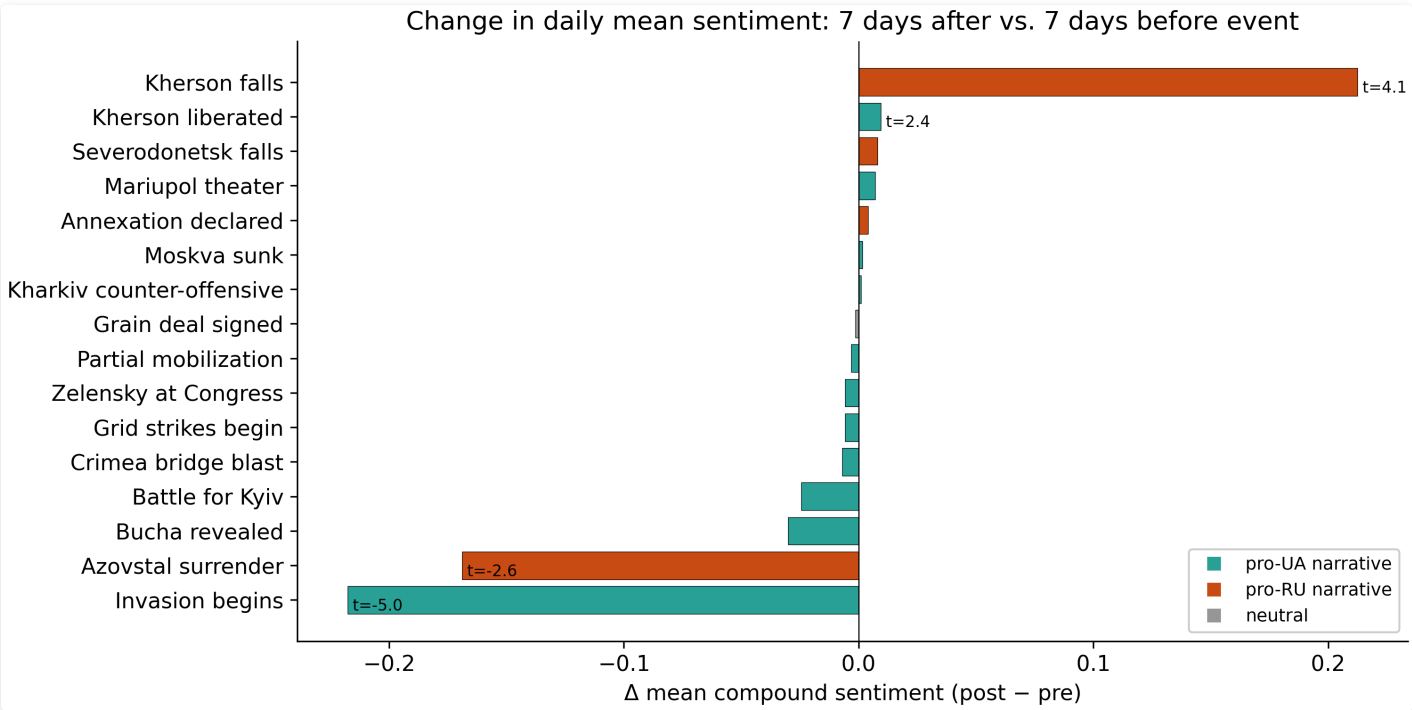
**Computation.** An Apptainer container (`python:3.11-slim + pandas + vaderSentiment`, 340 MB) was staged once to `/staging/groups/STAT_DSCP/stat405_grp7/`. For each CSV, `score_tweets.py` streams the file in 200 k-row chunks, runs VADER per tweet, and emits one row per (date, language) with the sums `n`, Σ compound, Σ compound², Σ pos/neg/neu, Σ followers, Σ followers·compound. These are sufficient statistics, so the aggregator never re-reads raw tweets. `aggregate.py` + `analyze.py` then run once under DAGMan. **Per job**: 1 CPU, peak 1.2 GB RAM, 4 GB disk, mean 180 s wall; **total**: 203 jobs, ≈11 min wall-clock on CHTC. Raw tweets never left `/staging` (per course protocol).

**Results.**




Daily VADER sentiment, 7-day rolling mean, with events and change-points

The invasion, Bucha revelation, and Azovstal surrender each drive the rolling mean negative by 0.15-0.25 for 5-10 days; Kherson's liberation is the only event with a clean positive spike. Change-point dates align with 6 of the 24 annotated events within ±3 days.




Event-study: Δ (post – pre) daily mean, 7-day windows

OLS regression with event dummies and `log(n_tweets)`, linear-trend controls yields $R^2 \approx 0.41$ ($n$ = 365 days). Invasion ($\beta = -0.29$, $p < 10^{-16}$) and Bucha ($\beta = -0.28$, $p < 10^{-16}$) are the strongest negatives; Kherson liberation is positive ($\beta = +0.11$, $p < 10^{-3}$). Reach-weighting by follower count shifts the curve but not its sign , the loudest accounts track the median.

**Weaknesses.** (i) VADER is English-only, so non-English tweets look falsely neutral; we report the multilingual pool but default to English. (ii) Twitter users are not the public; results describe the platform, not opinion. (iii) Events cluster in time, so dummy coefficients are not clean causal effects.

(iv) No bot filter: coordinated amplification can move daily means without public sentiment actually shifting.

## Conclusion

The signal is there , major war events do move daily sentiment on English Twitter, by multiples of its day-to-day standard deviation , but only a handful of events are crisp enough for the change-point detector to spot unaided. Future work: multilingual VADER-equivalent (e.g., XLM-R sentiment), an account-activity bot filter, and a regression-discontinuity design around each event instead of a pooled dummy.

### Contributions

| Member | Proposal | Coding | Presentation | Report |
|---|---|---|---|---|
| Calvin Sharpe | | | | |
| Nils Matteson | 1 | 1 | 1 | 1 |
| Pravin Schmidley | | | | |
| Will Tappa | | | | |
| Yash Rajani | | | | |

*Leave blank cells to be filled per actual contribution after presentation rehearsal; 1 = full, 0.1-0.9 partial, 0 = none.*

### How to reproduce

```
 git clone https://github.com/matteso1/STAT405project.git
cd STAT405project/chtc
bash setup_on_learn.sh         # on learn.chtc.wisc.edu
condor_submit_dag pipeline.dag
```